EXPERIMENT 1B

DEMONSTRATION OF JUPYTER WIDGETS AND JUPYTER AI APPLICATIONS

AIM

To demonstrate interactive widgets in a Jupyter Notebook and show a simple example of how an AI-related text-generation service can be called from a notebook when an authorised API key is available.

ALGORITHM

1. Open Jupyter Notebook and create a new Python 3 notebook.
2. Install the required widget package if it is not already installed.
3. Import **ipywidgets** and **IPython.display**.
4. Create a text box, button and output area.
5. Attach a click event to the button so the notebook displays a response based on the entered text.
6. For the AI demonstration, read an API key from an environment variable, send a short prompt to the chosen AI service, and display the response. Do not write API keys directly in notebook cells.

SOURCE CODE

```
# Part A: Interactive Jupyter widget
# Install once if required:
# %pip install ipywidgets

import ipywidgets as widgets
from IPython.display import display, clear_output

prompt_box = widgets.Text(
    value="Explain data visualization in one sentence.",
    description="Prompt:",
    layout=widgets.Layout(width="95%")
)
run_button = widgets.Button(description="Display")
output = widgets.Output()

def show_text(button):
    with output:
        clear_output()
        print("Entered prompt:", prompt_box.value)
        print("Widget demonstration completed.")

run_button.on_click(show_text)
display(prompt_box, run_button, output)


# Part B: Optional AI API example
# Requires an account/API key for the selected provider.
# Keep the key in an environment variable; never hard-code it.
#
# import os
# from openai import OpenAI
# client = OpenAI(api_key=os.environ["OPENAI_API_KEY"])
# response = client.responses.create(
#     model="MODEL_AVAILABLE_TO_YOUR_ACCOUNT",
#     input="Explain data visualization in one sentence."
# )
# print(response.output_text)
```

OUTPUT

```
Entered prompt: Explain data visualization in one sentence.
Widget demonstration completed.

For the optional AI section, the displayed response depends on the provider,
model access and API key configured by the user.
```

RESULT

An interactive text widget and button were demonstrated in Jupyter Notebook. The optional AI API code shows how a notebook can send a prompt to an authorised AI service and display its response when valid credentials and an available model are configured.